

Memo 03: MP1 Mass-Conservation Check and Updated Diagnosis

FVCOM-MP Delaware Test Cases

May 1, 2026

Purpose

This memo updates the MP1 mass-conservation diagnosis after rerunning cases **b** and **c** with the same sediment input now used by case **a**. The present build still bypasses the sediment horizontal `adv_scal_backward` and `fct_sed` block in `mod_sed.F`. The new result is a clean plastic split-path test: cases **b** and **c** now evolve almost identically, track case **a** for about three days, and then diverge together onto an oscillatory, lower-mass branch.

Archived Analysis Outputs

Six mass-conservation snapshots are now preserved under `FVCOM_MP_test_run/PYTHON/output/mass_conservation/archive/`.

Version	Archive label	Meaning
v001	v001_before_case_a_sediment_on	old comparison with case a using <code>SEDIMENT_MODEL=F</code>
v002	v002_case_a_sediment_on_full10d	case a with <code>SEDIMENT_MODEL=T</code> and the original sediment BBL settings
v003	v003_case_a_sediment_on_no_bbl	case- a diagnostic with <code>SEDIMENT_MODEL=T</code> and sediment BBL roughness switches disabled
v004	v004_sed_start_delayed	case a with sediment initialized but active <code>Advance_Sed</code> delayed beyond the run
v005	v005_sed_no_adv_fct	case a with active sediment runtime, but sediment <code>adv_scal_backward/fct_sed</code> bypassed
v006	v006_b_c_mirrored_sediment	cases b/c rerun with the same sediment input as case a ; all cases use <code>mp_sediment_a_no_adv.inp</code>

The live mass-conservation output folder has also been regenerated from the v006 state. The archived files include `mass_summary.csv`, `mass_timeseries_long.csv`, and all current comparison plots.

Case-a Evidence Trail

All archived comparisons cover the same FVCOM model-day window, 58485.0–58494.957031, or comparison day 0.00–9.96.

The two low-mass runs are v002 and v003, both of which execute active sediment horizontal advection and FCT limiting. The two high-mass controls are v001 and v004: either sediment is

Version	Case-a sediment treatment	Final total (kg)	Final vs. v001 (kg)
v001	sediment model off	1001.25	0.00
v002	sediment on, original BBL settings	582.41	-418.85
v003	sediment on, BBL disabled	573.28	-427.97
v004	sediment initialized, <code>Advance_Sed</code> delayed	1001.25	0.00
v005	sediment on, <code>adv_scal_backward/fct_sed</code> by-passed	994.20	-7.05

absent, or sediment setup is present but active `Advance_Sed` returns before runtime sediment transport. The new v005 result is the key isolation: it keeps active sediment runtime and starts sediment immediately (`SED_START` = 1), but bypasses the sediment horizontal advection/FCT block. Case a then recovers to 994.20 kg, within 7.05 kg of the old v001/v004 level.

Current v006 Output Summary

Case	Current output state	Final water (kg)	Final bed (kg)	Final total (kg)
a	rerun with sediment advection/FCT by-passed	994.20	0.00	994.20
b	mirrored sediment input, <code>PLAST_FLOC=T</code>	347.56	0.00	347.56
c	mirrored sediment input, <code>PLAST_FLOC=T</code>	348.52	0.00	348.52

The hard concentration cap remains inactive. The largest saved MP1 concentration in the v006 live analysis is 4.37×10^{-4} kg m⁻³ for case a, 2.38×10^{-4} kg m⁻³ for case b, and 2.38×10^{-4} kg m⁻³ for case c; all cap counters are zero. The saved split fields are exactly internally consistent in the new outputs: `mp1_agg` + `mp1_dis` = `mp1` to the saved precision for both b and c. The aggregated component remains zero throughout both runs; all water-column mass is in the disaggregated `mp1_dis` component.

v006 Divergence Timing

Cases b and c are now effectively the same experiment. Their maximum total-mass separation over the full run is only 1.68 kg, and their final totals differ by 0.96 kg. This confirms that the old b/c separation was driven by the different sediment files, not by a separate plastic behavior.

Comparison day	Case a (kg)	Case b (kg)	Case c (kg)	a-b (kg)
1.00	20.13	20.13	20.13	0.00
2.00	73.25	73.25	73.25	0.00
3.00	147.34	141.87	141.87	5.48
4.00	237.41	181.30	181.32	56.11
5.00	371.85	222.70	223.07	149.15
9.96	994.20	347.56	348.52	646.64

The first nontrivial a/b split occurs just before day 3: $|a-b| > 1$ kg at day 2.957031, $|a-b| > 5$ kg at day 3.000000, and $|a-b| > 50$ kg at day 4.000000. After day 3, case a has no sign changes in its step-to-step total-mass tendency, while cases b and c each have 28 sign changes. Thus the new

failure mode is not a smooth offset; it is an oscillatory split-branch response that starts once the plume has evolved for about three days.

Relevant Call Trace

1. **Setup order.** In `fvcom.F:643--651`, `SETUP_SED` is called before `SETUP_PLAST`.
2. **Timestep order.** In `internal_step.F:1092--1137`, `ADVANCE_SED` is called before `ADVANCE_PLAST`. Plastic therefore sees whatever state sediment leaves behind each internal step.
3. **Sediment startup guard.** In `mod_sed.F:883`, `Advance_Sed` returns immediately when `iint < sed_start`. The v004 test used this to keep sediment setup active while removing active sediment runtime.
4. **Bypassed block in the current build.** In `mod_sed.F:1016--1033`, the sediment `adv_scal_backward` call and `fct_sed` limiter are commented, while `sed(i)%cnew = sed(i)%conc` and `sed(i)%cflx = 0.0_sp` keep the rest of `Advance_Sed` moving.
5. **Shared scalar machinery.** `Adv_Scal_Backward` begins at `mod_scal.F:648`. During that routine, bottom scalar-gradient arrays `PFPXB/PFPYB` are written at `mod_scal.F:911--912`. This remains a concrete shared-state candidate, although the current test only proves that the sediment `adv_scal_backward/fct_sed` stage is implicated, not which line inside the stage is responsible.
6. **Split plastic branch.** In the `PLAST_FLOC=T` path, `mod_plast.F:1166--1186` transports `conc_a` and `conc_d` with separate `adv_scal` calls. It then applies vertical diffusion separately at `mod_plast.F:1203--1204`, applies split OBC/source handling at `mod_plast.F:1543--1605`, clamps `cnew_a/cnew_d` at `mod_plast.F:1657--1660`, and finally recombines at `mod_plast.F:1709--1716`. In the new v006 runs, `SOURCE_AG_RATE=0` and `OBC_AG_RATE=0`, so this branch should reduce to a disaggregated-only version of the no-floc case.
7. **Symmetry check.** At the direct call-signature level, `conc_d` is close to the no-floc `conc` path: `adv_scal(conc_d,cnew_d,...)` uses $cdis \times (1 - A_{rate_s0})$, which equals `cdis` when `SOURCE_AG_RATE=0`; the matching no-floc call is `adv_scal(conc,cnew,...)` at `mod_plast.F:1261--1267`. However, the full operator path is not directly symmetric because the split branch still executes floc-rate setup, split deposition/erosion code, separate zero-field aggregate transport, split OBC/source calls, split clamping, split MPI exchanges, and a later recombination step.

Diagnosis

1. **The problem is no longer just “sediment runtime on”.** v002 and v003 showed that active sediment runtime could move case-a final domain MP1 mass from about 1001 kg down to 573–582 kg. v004 then showed this was not caused by sediment setup, sediment namelist presence, or the sediment-enabled river forcing path, because delaying active `Advance_Sed` recovered the old mass curve.

2. **The new smoking gun is the sediment horizontal advection/FCT stage.** v005 keeps active sediment runtime, starts sediment immediately, and keeps the non-advection parts of `Advance_Sed` active. Closing only the sediment `adv_scal_backward/fct_sed` block recovers the v001/v004-like case-a result. That makes this block the leading source of the unwanted sediment/plastic interaction.
3. **v006 moves the active problem to the plastic split path.** With the sediment input now mirrored, cases b and c are nearly identical. The old b/c difference was therefore a sediment-input difference, not a separate plastic mechanism. The remaining large a versus b/c split is now controlled by `PLAST_FLOC`: case a uses the combined no-floc concentration, while cases b/c use split `conc_a/conc_d` fields.
4. **The effect is not active aggregation, bed exchange, or clipping.** In v006, the aggregated component is zero throughout cases b/c, the bed mass is zero, deposition/erosion exchange is zero, split-minus-combined mismatch is zero, and cap counters are zero. The oscillatory mass loss therefore occurs while the floc branch is effectively a disaggregated-only split transport problem.
5. **`conc_d` is not numerically identical to `conc`.** The physics expectation is correct: with `conc_a=0` and no kinetic transfer, `conc_d` should behave like the no-floc combined `conc`. The implementation does not enforce that equivalence. In case b, `conc_d` passes through the split-only path, including floc-rate calculations at `mod_plast.F:1021--1044`, split settling at `mod_plast.F:2729--2820`, split erosion/surface-flux updates at `mod_plast.F:2383--2440`, split OBC/source calls at `mod_plast.F:1543--1605`, and split clamp/exchange/recombine logic at `mod_plast.F:1657--1716`. Case a uses the combined branches for the same conceptual operators. Therefore the v006 difference can arise from numerical/operator asymmetry even when aggregate physics is inactive.
6. **The day-3 timing points toward a transport or boundary trigger.** Cases a, b, and c are indistinguishable through the first two days and begin separating just before day 3. After that, the no-floc case increases smoothly, while the split cases alternate positive and negative mass tendencies. The most likely interpretation is that the split branch begins interacting differently with a transport feature, open-boundary condition, wet/dry edge, or strong-gradient region once the plastic plume reaches it.
7. **BBL roughness stays demoted.** The v003 BBL-off run was only 9.13 kg lower than the v002 BBL-on run, while both were more than 410 kg lower than v005. BBL roughness is not the leading explanation for the present case-a mass shift.
8. **Primary hypothetical cause.** The strongest current hypothesis is that the `PLAST_FLOC=T` split path is not algebraically equivalent to the no-floc combined path when `SOURCE_AG_RATE=0`, `OBC_AG_RATE=0`, and aggregation remains zero. The problem may be in one of five nearby places: floc-rate setup before the normal update, split deposition/erosion handling, the extra zero-field `adv_scal/ vdif_scal` calls for `conc_a`, split OBC/source handling, or the final clamp/exchange/recombine order. The oscillation makes the OBC/source or halo-exchange stages especially plausible, but this still needs an operator-by-operator mass checkpoint.

Suggested Next Move

Keep v006 as the clean comparison baseline and do one short, high-instrumentation rerun around the divergence window. The goal is to find the first operator where **b** departs from **a**.

1. **Add stage mass checkpoints for case a and case b.** For days 2.75–3.25, write domain-integrated MP1 mass after each plastic operator: start of `Advance_Plast`, after deposition, after erosion/update-bottom, after horizontal advection, after vertical diffusion, after OBC, after point-source BC, after nonnegative clamp, after MPI exchange, and after recombination. For **b**, log `conc_a`, `conc_d`, and `conc_a + conc_d` separately.
2. **Run a null-split equivalence test.** Add a temporary diagnostic switch so `PLAST_FLOC=T` allocates and outputs split fields, but skips all aggregate work and transports only `conc_d` through the exact combined no-floc operator. This means no floc-rate setup, no zero `conc_a` transport, no split OBC/source calls, no split cap, and no split exchange/recombine order. If this matches case **a**, the remaining problem is inside the split operator sequence rather than setup or output.
3. **Bisect the split operator sequence.** Starting from the current **b** setup, test these small changes one at a time: skip the zero `conc_a` `adv_scal/vdif_scal` calls when aggregate rates are zero; apply OBC/source only to `conc_d`; then force the floc-branch final clamp/exchange/recombine order to mimic the no-floc branch as closely as possible.
4. **Add an open-boundary/source budget.** Accumulate MP1 source input and open-boundary export for the combined field in case **a** and for `conc_a + conc_d` in case **b**. If the day-3 split coincides with a jump in export, the OBC path becomes the leading target.
5. **Check serial versus parallel behavior.** If the oscillation changes with processor count, focus immediately on the split exchange calls around `mod_plast.F:1512--1516` and `mod_plast.F:1679--1716`.